

Ben McDowell

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EDUCATION

University of Bath

Bath, UK

University of Bath - B.Sc. (Hons) Computer Science and AI (Predicted First Class Honours) Sep. 2022 – Jul 2026

The College of Richard Collyer

Horsham, UK

*Computer Science, Extended Project Qualification (EPQ), French, Maths (A*A*AA)* Sep. 2020 – Jun 2022

EXPERIENCE

Software Engineer Intern

Jun. 2025 – Aug 2025

Jefferies

London, UK

- Deployment Automation Tool – Designed and delivered a full-stack (Angular, C# WebAPI, SQL Server) CI/CD tool, integrating with multiple APIs to automate change ticket management, credential retrieval, and application deployment. Implemented role-based access controls, real-time monitoring, and audit logging. Onboarded Treasury Technology team, reducing deployment time by 60% (saving 1.5+ hours per week).
- SSO Migration – Led Treasury Technology team's first migration to Microsoft Entra ID (SSO), strengthening security, enabling mobile access for stakeholders, and authoring setup documentation to support future integrations.

Business Intelligence Engineer Intern

Jun. 2024 – Jun 2025

Amazon

London, UK

- Developed an alternative forecasting model, improving average accuracy by 35% across all EU 3P carriers, supporting better decision-making during High Volume Events.
- Built a complete ETL pipeline and a daily-refreshing QuickSight dashboard to track a +\$30M plan for all EU 3P carriers, enabling granular cost visibility and proactive management of forecasted headwinds.
- Delivered a tool for carrier managers to dynamically steer volume between 1P and 3P carriers during capacity constraints, facilitating rapid 3P flex-ups at a regional level.
- Created a Python-based tool deployed with AWS Lambda to automate forecast input during High Volume Events, improving data consistency, and enabling historical trend analysis across multiple dashboards.

Data Analyst Intern (Part-time)

Nov. 2022 – Apr. 2024

Play Sports Network

Bath, UK

- Produced commercial reports in Looker Studio and delivered them to company partners across 23 social channels and various platforms. Alongside analysis of social media metrics using SQL (BigQuery), Python and Excel.

TECHNICAL SKILLS

Languages: Python, SQL, C, C#, Java, JavaScript, HTML/CSS, Bash

Frameworks: .NET, Angular, React, Flask

Developer Tools: AWS, Git, Docker, Terraform, Google Cloud Platform, Visual Studio, CyberArk Vault, JFrog Artifactory, Atlassian Suite, ServiceNow, REST APIs (Postman), GitHub Actions

Libraries: pandas, NumPy, Matplotlib, scikit-learn, PyTorch

Certifications: AWS Certified Cloud Practitioner

PROJECTS

Personal Informatics System | *Python, Flask, React, GitHub Actions*

Sep. 2023 – May 2024

- Designed and implemented a full-stack mobile application leveraging a pre-trained OpenAI classification model to automate waste categorization from photos.
- Developed RESTful APIs in Flask and a responsive React frontend; ensured robustness with comprehensive unit tests and CI/CD pipelines.
- Led Agile workflow and iterative delivery as Scrum Master and Product Owner for an 8-person team.

Email spam classifier using Naive Bayes | *Python, numpy*

Jan. 2023 – Mar 2023

- Implemented a Naive Bayes classifier from scratch using numpy, achieving over 90% accuracy on a dataset of 900 emails. Applied Laplace smoothing to address zero-frequency problems and validated model performance through cross-validation.